

Task 1: Java Basics & OOPs Fundamentals

Goal:

Understand Java basics and object-oriented programming concepts.

Requirements:

- Write simple Java programs (loops, conditions, arrays)
- Implement classes, objects, inheritance, and polymorphism
- Use encapsulation with getters and setters

Code Snippets:

```
class Person {  
    private String name;  
    private int age;  
    public void setName(String name) {  
        this.name = name;  
    }  
    public String getName() {  
        return name;  
    }  
    public void setAge(int age) {  
        this.age = age;  
    }  
    public int getAge() {  
        return age;  
    }  
}
```

```

    }

    public void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }
}

class Student extends Person {
    private int marks;

    public void setMarks(int marks) {
        this.marks = marks;
    }

    public int getMarks() {
        return marks;
    }

    @Override
    public void displayInfo() {
        System.out.println("--- Student Details ---");
        System.out.println("Name: " + getName());
        System.out.println("Age: " + getAge());
        System.out.println("Marks: " + marks);
    }
}

public class task1 {
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.setName("Mandar");
        s1.setAge(19);
    }
}

```

```

s1.setMarks(95);
s1.displayInfo();
System.out.println("\nPass or Fail ?");
if (s1.getMarks() >= 40) {
    System.out.println("Result: Pass");
} else {
    System.out.println("Result: Fail");
}
System.out.println("\nNumbers from 1 to 5 using loop:");
for (int i = 1; i <= 5; i++) {
    System.out.println(i);
}
System.out.println("\nArray:");
int[] numbers = {10, 20, 30, 40, 50};
int sum = 0;
for(int num: numbers){
    System.out.print(num+" ");
}
for (int num : numbers) {
    sum += num;
}
System.out.println("\nSum of array elements: " + sum);
}
}

```

Screenshots:

```
J task1.java X
J task1.java > ...

1
2  class Person {
3      private String name;
4      private int age;
5
6      public void setName(String name) {
7          this.name = name;
8      }
9
10     public String getName() {
11         return name;
12     }
13
14     public void setAge(int age) {
15         this.age = age;
16     }
17
18     public int getAge() {
19         return age;
20     }
21
22     public void displayInfo() {
23         System.out.println("Name: " + name);
24         System.out.println("Age: " + age);
25     }
26 }
27
28  class Student extends Person {
29      private int marks;
30
31      public void setMarks(int marks) {
32          this.marks = marks;
33      }
34
35      public int getMarks() {
36
37          public int getMarks() {
38              return marks;
39          }
40
41          @Override
42          public void displayInfo() {
43              System.out.println(x: "--- Student Details ---");
44              System.out.println("Name: " + getName());
45              System.out.println("Age: " + getAge());
46              System.out.println("Marks: " + marks);
47          }
48      }
49  }
50
51  public class task1 {
52      Run main | Debug main | Run | Debug
53      public static void main(String[] args) {
54
55          Student s1 = new Student();
56
57          s1.setName(name: "Mandar");
58          s1.setAge(age: 19);
59          s1.setMarks(marks: 95);
60
61          s1.displayInfo();
62
63          System.out.println(x: "\nPass or Fail ?");
64
65          if (s1.getMarks() >= 40) {
66              System.out.println(x: "Result: Pass");
67          } else {
68              System.out.println(x: "Result: Fail");
69          }
70      }
71  }
```

```
J task1.java X
J task1.java > ...
48 public class task1 {
49     public static void main(String[] args) {
64         System.out.println(x: "Result: Fail");
65     }
66
67     System.out.println(x: "\nNumbers from 1 to 5 using loop:");
68
69     for (int i = 1; i <= 5; i++) {
70         System.out.println(i);
71     }
72
73
74     System.out.println(x: "\nArray:");
75
76     int[] numbers = {10, 20, 30, 40, 50};
77
78     int sum = 0;
79     for(int num: numbers){
80         System.out.print(num+" ");
81     }
82
83     for (int num : numbers) {
84         sum += num;
85     }
86
87     System.out.println("\nSum of array elements: " + sum);
88
89
90     }
91 }
```

Output:

```
• --- Student Details ---
Name: Mandar
Age: 19
Marks: 95

Pass or Fail ? :
Result: Pass

Numbers from 1 to 5 using loop:
1
2
3
4
5

Array:
10 20 30 40 50
Sum of array elements: 150
```

Commands Used

```
javac task1.java
```

```
java task1
```

Explanation

- **Class and Object:** Person and Student are classes, and s1 is an object of the Student class.
- **Encapsulation:** Data members are declared private and accessed using getter and setter methods.
- **Inheritance:** The Student class inherits properties and methods from the Person class using extends.
- **Polymorphism:** The displayInfo() method is overridden in the Student class.
- **Loops:** for loops are used to print numbers and process array elements.
- **Conditions:** if-else statements are used to check pass or fail status.
- **Arrays:** An integer array is used to store and display multiple values.

GitHub repo link :

<https://github.com/MandarJoshi12/Alfido-Tech-Internship-Tasks.git>

Hosted Link :

https://onlinegdb.com/pdECUZ_ol

